Do Now · retrieval

Lesson 106 · What an Operating System Does

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Distinguish RAM from virtual storage.

Q2

In a BST, where is a value less than the current node placed?

Q3

Why can an array provide direct indexed access?

Q4

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Manage processes — The OS creates and schedules processes; programs share CPU time.

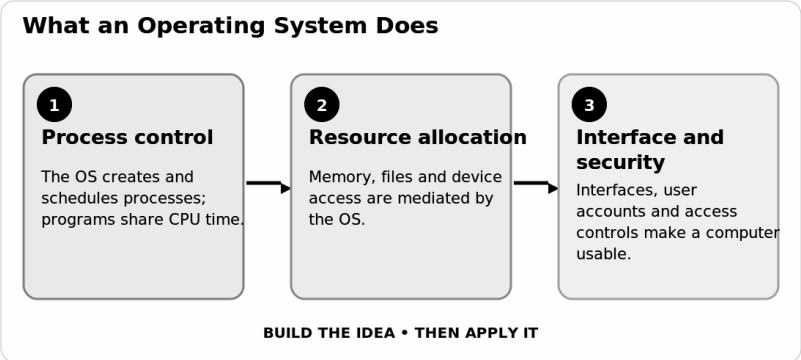
Core idea 2

Manage resources — Memory, files and device access are mediated by the OS.

Core idea 3

Provide services — Interfaces, user accounts and access controls make a computer usable.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

operating system — System software managing hardware resources and services.

process — An executing instance of a program.

device driver — Software allowing the OS to communicate with hardware.

spooling — Queuing data such as print jobs for a slower peripheral.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

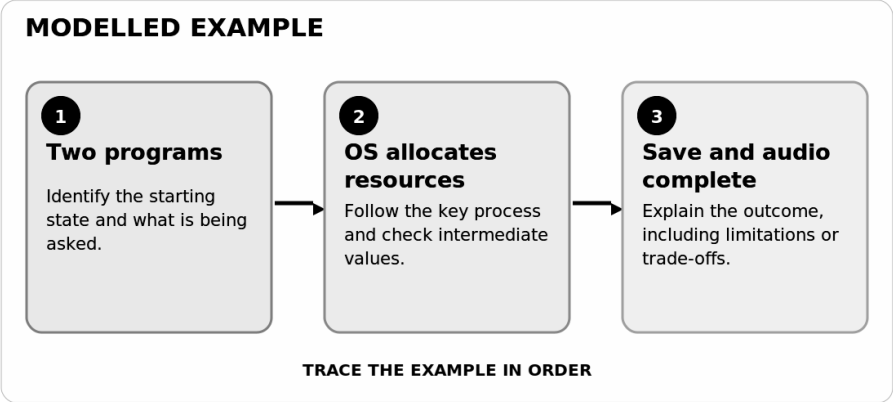
Problem

A word processor saves a document while music is playing. Explain three OS services involved.

Follow the worked method

1. Model answer — The OS allocates CPU time to both processes; the file system writes the document to storage; a device driver controls sound output. Memory management keeps process data protected. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

Question 1

Describe three functions of an operating system, linking each to a user action. [6]

Question 2

A computer has three applications requesting the printer. Explain how the OS can handle this. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

Question 3

Evaluate the claim that application programs can directly manage all hardware without system software. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The operating system only provides a graphical user interface.

Use key vocabulary from memory

1. Explain operating system in your own words.

2. Explain process in your own words.

3. Explain device driver in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.